Team Members :

Shivam Mishra(S24CSEU1470) &

Priyanshu Prakash Singh(S24CSEU1462)

BlockChit :

Decentralized and community Lending Platform

PROJECT ABSTRACT:

BlockChit is a blockchain-powered decentralized application created to digitize and secure community-based savings and lending practices that are traditionally managed through informal trust systems. The core intention behind this project is to replace manual bookkeeping, verbal agreements, and centralized control with a transparent, rule-driven digital platform where every financial action is recorded on a public ledger. This approach reduces dependency on intermediaries and significantly lowers the chances of fraud, bias, or financial disputes among members.

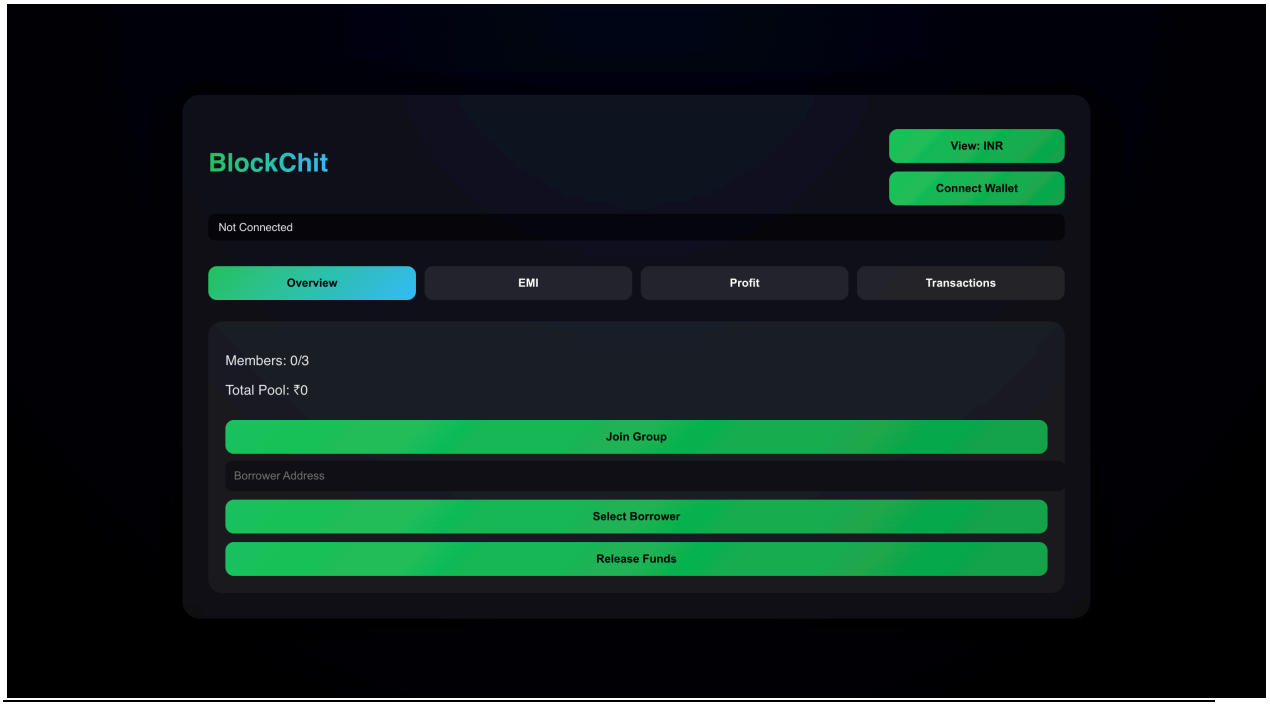
The problem addressed by BlockChit arises from common issues in conventional chit funds and rotating savings groups such as delayed payments, mismanagement of pooled funds, absence of accountability, and lack of real-time visibility. The proposed system introduces a structured digital model where a fixed number of participants contribute equal amounts into a

shared pool governed entirely by smart contracts. Borrower selection, fund release, installment schedules, and profit distribution are executed automatically based on predefined logic, ensuring fairness and eliminating human interference in financial decisions.

From a technological perspective, BlockChit is built using Solidity smart contracts deployed on the Ethereum blockchain, while the user interface is developed with React to provide a responsive and interactive experience. Wallet integration through MetaMask enables secure identity verification and direct transaction execution without exposing sensitive credentials. To enhance accessibility, the application presents financial values in both cryptocurrency (ETH) and an approximate INR conversion, allowing users with minimal blockchain knowledge to understand their financial position clearly.

Testing and practical demonstrations indicate that BlockChit operates reliably in handling group creation, automated loan issuance, EMI enforcement with time locks, and equitable profit sharing. The integrated on-chain transaction history offers complete auditability, empowering users to verify every contribution and withdrawal independently.

Overall, BlockChit demonstrates how decentralized technologies can be applied beyond cryptocurrencies into real-world cooperative finance systems. By combining automation, transparency, and security, the platform establishes a trust-independent financial environment and highlights the potential of blockchain solutions in reshaping grassroots economic collaboration.


BlockChit
View: INR
Connect Wallet
Not Connected
Overview
EMI
Profit
Transactions
Members: 0/3
Total Pool: ₹0
Join Group
Borrower Address
Select Borrower
Release Funds